

CHE 311 ProblemSet #2

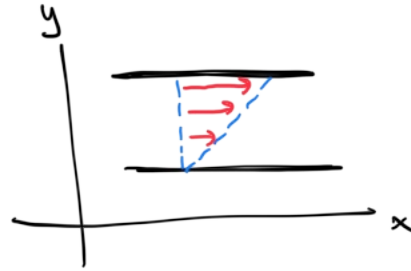
Problem 1B.1

a.) $v_x = by, v_y = 0, v_z = 0$

Gradients: $\frac{\partial v_x}{\partial y} = b$; all others = 0

Stress: $e_{xy} = e_{yx} = \frac{1}{2}b \rightarrow \tau_{xy} = \tau_{yx} = \mu b$

$$\rho \mathbf{v} \otimes \mathbf{v} = \rho \begin{pmatrix} b^2 y^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$



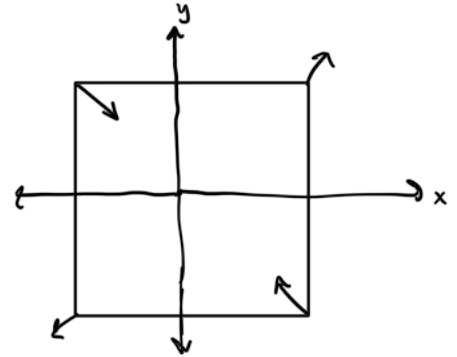
B.) $v_x = by, v_y = bx, v_z = 0$

Gradients: $\frac{\partial v_x}{\partial y} = b, \frac{\partial v_y}{\partial x} = b$

Stress: $e_{xy} = e_{yx} = \frac{1}{2}(b+b) = b \rightarrow \tau_{xy} = \tau_{yx} = 2\mu b$

$e_{xx} = e_{yy} = 0 \rightarrow \tau_{xx} = \tau_{yy} = 0$

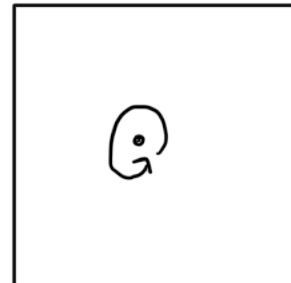
$$\rho \mathbf{v} \otimes \mathbf{v} = \rho \begin{pmatrix} b^2 y^2 & b^2 xy & 0 \\ b^2 xy & b^2 x^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$



C.) $v_x = -by, v_y = bx, v_z = 0$

Gradients: $\frac{\partial v_x}{\partial y} = -b, \frac{\partial v_y}{\partial x} = b$

$$\rho \mathbf{v} \otimes \mathbf{v} = \rho \begin{pmatrix} b^2 y^2 & -b^2 xy & 0 \\ -b^2 xy & b^2 x^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$



Rigid body
Rotation

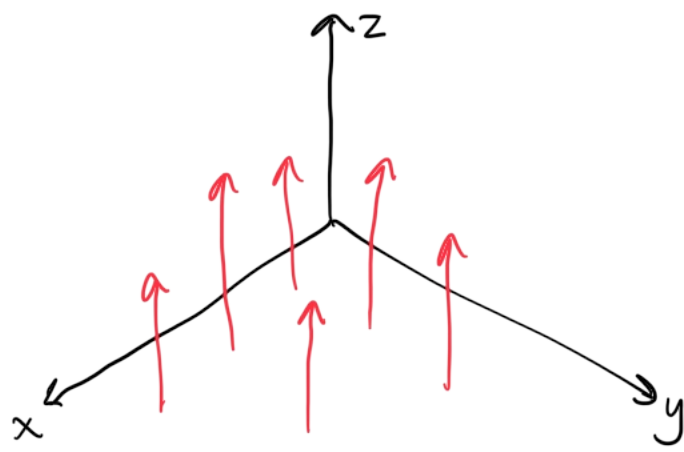
$$d.) \quad v_x = -\frac{1}{2}bx \quad v_y = -\frac{1}{2}by \quad v_z = bz$$

$$\text{gradients: } \frac{\partial v_x}{\partial x} = -\frac{1}{2}b \quad \frac{\partial v_y}{\partial y} = -\frac{1}{2}b \quad \frac{\partial v_z}{\partial z} = b$$

$$\text{Stress: } e_{xx} = -\frac{1}{2}b \quad e_{yy} = -\frac{1}{2}b \quad e_{zz} = b$$

$$\tau_{xx} = -\mu b \quad \tau_{yy} = -\mu b \quad \tau_{zz} = 2\mu b$$

$$p(\mathbf{r}) = p \begin{pmatrix} \frac{b^2}{4}x^2 & \frac{b^2}{4}xy & -\frac{b^2}{2}xz \\ \frac{b^2}{4}xy & \frac{b^2}{4}y^2 & -\frac{b^2}{2}yz \\ -\frac{b^2}{2}xz & -\frac{b^2}{2}yz & b^2z^2 \end{pmatrix}$$



2B.2

$$\frac{d\tau_{xz}}{dx} + \rho g = 0$$

$$\tau_{xz} = \mu \frac{dv_z}{dx}$$

↓

$$\frac{d}{dx} \left(\mu \frac{dv_z}{dx} \right) = -\rho g \rightarrow \mu \frac{dv_z}{dx} = -\rho g x + C_1$$

$$\tau_{xz}(x=\delta) = 0 \rightarrow C_1 = \rho g \delta$$

$$\frac{dv_z}{dx} = \frac{\rho g}{\mu} (\delta - x)$$

$$v_z(x) = \frac{\rho g}{\mu} \left(\delta x - \frac{1}{2}x^2 \right)$$

$$v_{max} = v_z(\delta) = \frac{\rho g \delta^2}{2\mu}$$

$$\Gamma = \bar{v} \delta = \frac{\rho g \delta^3}{3\mu}$$

$$\bar{v} = \frac{1}{\delta} \int_0^\delta v dx = \frac{\rho g \delta^2}{3\mu}$$

2B.3

$$\frac{d\tau_{xz}}{dx} = \frac{dp}{dz} = \frac{p_i - p_o}{L} = -G$$

$$\tau_{xz} = -Gx$$

$$\mu \frac{dv_z}{dx} = \tau_{xz} = -Gx \rightarrow \frac{dv_z}{dx} = -\frac{G}{\mu} x$$

$$v_z(x) = \frac{G}{2\mu} (B^2 - x^2) = \frac{(p_o - p_i) B^2}{2\mu L} \left[1 - \left(\frac{x}{B} \right)^2 \right]$$

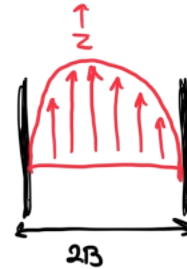
$$v_{max} = v(0) = \frac{GB^2}{2\mu}$$

$$\bar{v} = \frac{1}{2B} \int_{-B}^B v dx = \frac{GB^2}{3\mu}$$

$$\bar{v}/v_{max} = \frac{2}{3}$$

$$Q = W \int_{-B}^B v dx = W \cdot 2 \int_0^B \frac{G}{2\mu} (B^2 - x^2) dx = \frac{2WB^3 G}{3\mu}$$

$$Q = \frac{2WB^3}{3\mu} \frac{p_o - p_i}{L}$$



2B.7

momentum balance: $\frac{1}{r} \frac{d}{dr} (r \tau_{rz}) = 0 \rightarrow r \tau_{rz} = 0$

$$\text{Newton's Law: } \tau_{rz} = \mu \frac{dv_z}{dr} \rightarrow \mu r \frac{dv_z}{dr} = C \rightarrow \frac{dv_z}{dr} = \frac{C}{\mu r}$$

$$v_z(r) = \frac{C}{\mu} \ln r + C_2$$

$$\text{Boundary: } v_z(KR) = v_o \quad v_z(R) = 0$$

$$\frac{v_z(r)}{v_o} = \frac{\ln(r/R)}{\ln K}$$

$$D.) \epsilon = 1 - K \ll 1$$

$$F = \frac{2\pi L \mu v_o}{\epsilon} \left(1 - \frac{1}{2}\epsilon - \frac{1}{8}\epsilon^2 + \dots \right)$$

$$B.) W = \int_{KR}^R p v_z(r) 2\pi r dr = 2\pi p \frac{v_o}{\ln K} \int_{KR}^R r \ln \frac{r}{R} dr$$

$$W = \frac{\pi p v_o R^2}{2} \left[\frac{1-K^2}{\ln(1/K)} - 2K^2 \right]$$

$$C.) F = \int_0^{2\pi} \tau_{rz}(KR) L d\theta = 2\pi(KR) L \tau_{rz} \Big|_{r=KR}$$

$$\tau_{rz} = \frac{dv_z}{dr} = \mu \frac{v_o}{\ln K} \frac{1}{r}$$

$$F = \frac{2\pi L \mu v_o}{\ln(1/K)}$$



2B.8

$$T = 20^\circ\text{C} \quad \mu = 1.002 \times 10^{-3} \quad \rho_w = 998.2 \quad \rho_m = 1.594 \quad d = 0.01 \quad L = 120 \quad H = 1.0$$

$$\Delta P = (\rho_m - \rho_w) g H$$

$$\rho_m - \rho_w = (1.594 - 0.9982) \text{ g/cm}^3 = 595.8 \text{ kg/m}^3$$

$$H = 1 \text{ in} = 0.0254 \text{ m}$$

$$\Delta P = 595.8 \cdot 9.80 \cdot 0.0254 = 1.484 \times 10^2 \text{ Pa}$$

$$\bar{V} = \frac{d^2}{32\mu L} \Delta P, \quad Q = \bar{V} \cdot \frac{\pi d^2}{4}$$

$$\bar{V} = \frac{(0.54 \times 10^{-4})(1.484 \times 10^2)}{32(1.002 \times 10^{-3})(3.048)} = 9.80 \times 10^{-5} \text{ m/s}$$

$$Q = \bar{V} \cdot \frac{\pi d^2}{4} = 4.96 \times 10^{-12} \text{ m}^3/\text{s}$$

Mass Flow

$$\dot{m} = \rho_w Q = (4.96 \times 10^{-12})(998.2) = 4.96 \times 10^{-9} \text{ kg/s}$$

$$\dot{m} = \dot{m}(3600)(0.20462) = 3.93 \times 10^{-5} \text{ lb/hr}$$

